Day 1 Student Handout

Geometry Summer Credit | Coordinate Geometry and Angle Relationships

Name:	Date:	Period:

Learning Targets

I can use coordinates to find distances and midpoints. I can identify angle relationships and use equations to find missing angle measures.

Lesson 1A: Coordinate Plane, Distance, and Midpoint

Warm-Up

1.	Plot A(2, 3), B(2, -4), and C(-5, 3). Which two points make a horizontal segment? How do you know?

2.	Compute each value: |3 - (-4)|, |-5 - 2|, sqrt(49), sqrt(20), sqrt(72).

Concept Launch

Notice and Wonder

When two points make a diagonal segment, the horizontal change and vertical change can be treated like the legs of a right triangle.

3.	Plot P(-2, -1) and Q(4, 7). Draw the horizontal and vertical moves that connect the points. What are the two leg lengths?

Guided Examples

Problem	Work / notes
Find the distance between (1, 5) and (1, -2).	What changes? x or y? Distance:
Find the distance between (-3, 4) and (5, 4).	What changes? x or y?

	Distance:
Find the distance between (-2, -1) and (4, 7).	Horizontal change: Vertical change: Use a^2 + b^2 = c^2:
Find the midpoint of A(-6, 8) and B(2, -4).	Average the x-values: Average the y-values: Midpoint:

Practice

4.	Find the distance between (3, 2) and (3, -6).

5.	Find the distance between (-4, 1) and (6, 1).

6.	Find the distance between (0, 0) and (5, 12).

7.	Find the midpoint of (-8, 6) and (4, -2).

8.	Error analysis: A student says the distance from (0, 0) to (6, 8) is 14 because 6 + 8 = 14. Explain the mistake.

Day 1 Student Handout

Lesson 1B | Angle Vocabulary and Angle Relationships

Lesson 1B: Angle Vocabulary and Angle Relationships

Warm-Up

1.	Solve $x + 37 = 90$.

2.	Solve $3x - 8 = 112$.

3.	Two angles add to 180 degrees. One angle is 64 degrees. Find the other.

Key Vocabulary

Term	What it means / sketch
Complementary angles	
Supplementary angles	
Linear pair	
Vertical angles	
Adjacent angles	

Concept Launch

Look for Structure

When lines meet, some angle measures are connected because they form a straight line, make a right angle, or sit opposite each other.

4.	Sketch two intersecting lines. Label the four angles around the intersection. What angle pairs seem equal? What angle pairs seem to add to 180 degrees?

Guided Examples

Problem	Work / notes
Angles A and B are complementary. If mA = 28 degrees, find mB.	Equation: Solution:
Angles C and D form a linear pair. If mC = 5x + 10 and mD = 3x - 6, find x and both angle measures.	Equation: Solve for x: Angle measures:
Vertical angles are labeled 7x - 5 and 4x + 34. Find x and the angle measure.	Equation: Solve for x: Angle measure:

Practice

5.	Two angles are complementary. One angle is 41 degrees. Find the other angle.

6.	Two angles form a linear pair. One angle is 112 degrees. Find the other angle.

7.	Two vertical angles are labeled 9x + 4 and 6x + 31. Find x.

8.	Error analysis: A student says all adjacent angles are supplementary. Draw or describe a counterexample.

Reflection

9.	Which part of Day 1 felt most familiar? Which part do you want more practice with before the quiz?